

Mechanical Packing of Spherical Particles

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An idealized experimental study of particle packing was made. Spherical metal shot of several discrete, narrow size ranges was efficiently packed in glass containers by mechanical vibration. Packing arrangements and the dynamic process of packing were studied visually. One-size spheres packed in an orthorhombic arrangement with a density 62.5% of theoretical density. Forming of high-density multicomponent packings was shown to require at least a sevenfold difference between sphere sizes of the individual components. A quaternary packing with a density 95.1% of theoretical density was formed from spheres with diameter ratios 1:7:38:316 and volume compositions 6.1:10.2:23.0:60.7%, respectively. Such packings could be poured from their glass containers, thus proving that effective mechanical packing is simply an efficient arrangement of spheres of prescribed sizes and proportions. The significance and utility of this work to the ceramic and other industries is discussed.

I. Introduction

THE science of particle packing can be defined as the selection of proper sizes and proportions of particulate material so that larger voids are filled with smaller particles, and these voids are in turn filled with still smaller particles, and so on. No plastic deformation of the particles occurs, and it is possible to pour the material out of the container after packing. Packing of particles in an efficient arrangement is of basic importance to many branches of industry and science, and yet there is but meager published information on the subject, most of which was reported in the early 1930's. Because of the wide applicability of packing, there is a diversity in approach to the subject as represented by the concrete industry, makers of quality ceramic ware and refractories,^{1,2} packed tower and bed hydraulics engineering,^{3,4} industrial handling and packing,⁵ and soil geology.^{6,7} Most

of the data is empirical, being intended to solve specific industrial problems. With a single exception⁸ the literature appears to be devoid of reference to mechanical equipment or to methods of producing efficient packing. There are no textbooks devoted to the subject of particle packing. One book⁹ reviews most of the information before 1948 in a chapter.

About 1954 there was renewed interest in high-density packing of ceramic and metal particles owing to the impetus of atomic energy and space research. Most of these new studies are concerned with the packing of uranium oxide in long metal tubes by use of vibratory energy and have centered around experiments designed to determine optimum particle-size distributions for maximum packing density.

It is the writer's belief that the present investigation represents the first experimental and theoretical attempt to extend the study of idealized packing of spheres of different sizes initiated by Furnas,¹⁰ 1929 to 1931, and by Westman and Hugill¹¹ in 1930. Both of these investigations apparently were discontinued after revealing the importance of size ratios of the particle components for efficient packing.

II. One-Component Packings

Almost everyone who has experimented with packings has poured spherical particles of a single size into a container and applied tapping or tamping. Typically, the spheres constitute 60 to 64% of the volume of the packing (40 to 36% voids). Lead, sulfur, and steel shot, steel ball bearings, glass beads, rounded sand, round California beans, and poppy seeds all produce packings within this range. Results for a given material and container are reproducible to a fraction of a per cent. Sizes of the different materials tested varied considerably. The main conclusion is that neither material density nor a difference in size from one material to another significantly affects ultimate packed density.

For the present studies, axial vibration was used to produce particle packing. The influence of the conditions of vibration and the type of equipment on packing efficiency will be discussed in detail later. Nearly spherical steel shot, which was available in sizes down to about 100-mesh, was used for most of the packing components. By pouring the shot on a slightly inclined glass panel the roundest shot rolled down the panel and was collected for use; the more irregular shot remained on the panel and was discarded. Spherical tung-

Received January 23, 1961; revised copy received May 2, 1961.

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This work was supported by the U. S. Atomic Energy Commission under Contract AT(30-1)-2389, Development of Plutonium Bearing Fuel Materials.

¹ H. E. White and S. F. Walton, "Particle Packing and Particle Shape," *J. Am. Ceram. Soc.*, **20** [5] 155-66 (1937).

² W. O. Williamson, "Physical Relationships Between Clay and Water," *Trans. Brit. Ceram. Soc.*, **50** [1] 10-34 (1951); *Ceram. Abstr.*, **1952**, April, p. 75c.

³ M. Leva and M. Grummer, "Pressure Drop Through Packed Tubes: III, Prediction of Voids in Packed Tubes," *Chem. Eng. Progress*, **43** [12], *Trans. Am. Inst. Chem. Engrs.*, **713-18** (1947).

⁴ J. J. Martin, W. L. McCabe, and C. C. Monrad, "Pressure Drop Through Stacked Spheres; Effect of Orientation," *Chem. Eng. Progress*, **47** [2] 91-94 (1951).

⁵ D. R. Hudson, "Density and Packing in an Aggregate of Mixed Spheres," *J. Appl. Phys.*, **20**, 154-62 (1949).

⁶ L. C. Graton and H. J. Fraser, "Systematic Packing of Spheres, With Particular Relations to Porosity and Permeability: I-II," *J. Geol.*, **43** [8, Part 1] 785-909 (1935); *Ceram. Abstr.*, **15** [7] 221 (1936).

⁷ H. J. Fraser, "Experimental Study of Porosity and Permeability of Clastic Sediments," *J. Geol.*, **43**, 910-1010 (1935).

⁸ W. C. Bell, R. D. Dillender, H. R. Lominac, and E. G. Manning, "Vibratory Compacting of Metal and Ceramic Powders," *J. Am. Ceram. Soc.*, **38** [11] 396-404 (1955).

⁹ J. M. DallaValle, *Micromeritics, the Technology of Fine Particles*, 2d ed. Pitman Publishing Corp., New York, 1948. 555 pp.

¹⁰ C. C. Furnas, "Grading Aggregates: I, Mathematical Relations for Beds of Broken Solids of Maximum Density," *Ind. Eng. Chem.*, **23** [9] 1052-58 (1931); *Ceram. Abstr.*, **11** [4] 278 (1932).

¹¹ A. E. R. Westman and H. R. Hugill, "Packing of Particles," *J. Am. Ceram. Soc.*, **13** [10] 767-79 (1930).

Table I. Sphere Sizes Used for Mechanical Packing

For reference U. S. sieve mesh No.	Size opening (in.)	Metal shot used			Calculated average sphere diameter (in.)
		Steel	Tungsten	Aluminum	
6	0.132				
7	0.111	-6+7			0.124 (measured)
8	0.094				
10	0.079				
12	0.066				
14	0.056	-12+14			0.061
16	0.047	-14+16			0.052
18	0.039	-16+18			0.043
20	0.033	-18+20			0.036
25	0.028				
30	0.023	-25+30			0.026
35	0.020				
40	0.017	-35+40			0.019
45	0.014				
50	0.012	-45+50			0.013
60	0.010	-50+60			0.011
70	0.008				
80	0.007	-70+80			0.0075
100	0.006	-80+100			0.0065
200	0.003				
325	0.0017				
400	0.0015		-325 + 400	-325 + 400	0.0016
-400	<0.0015		-400	-400	<0.0016

sten powder (prepared by passing through a plasma-flame powder spraying torch) and atomized spherical aluminum metal powder were obtained for the fine components of packings. The thirteen different sizes of metal spheres used and their corresponding U. S. sieve mesh numbers are listed in Table I. Note that only "pure" components consisting of a single mesh size were used. This refinement increased the sensitivity of the experimental results, especially those relating to particle-size effects. Since each component passed through one sieve and was completely retained on the next smaller sieve, it will be convenient in this paper to refer to a particular component, e. g., -25+30 mesh as 30-mesh and -6+7 mesh as 7-mesh. Each mesh size is also designated by d , its average sphere diameter. For most of the components this size is the average of the two sieves used.

In Table II are presented the pycnometric measured densities of several of the single-sphere-size materials as well as their densities after packing in a glass graduate 1 in. in diameter. The intensity, frequency, and duration of vibration were not critical. Maximum packing density was rapidly attained and results were readily reproducible to $\pm 0.1\%$.

The inability of fine materials to pack as well as coarser materials is well known. Although the tungsten and aluminum were quite spherical and had smooth surfaces, other

forces associated with agglomeration came into play to destroy systematic packing.

Both the size and shape of the container used for packing influence the type and density of the packing. Apparently, in the previous work, the containers were sufficiently larger than the size of the particles so that boundary effects were negligible. By the use of transparent containers and a low-power stereoscopic microscope, it was possible in the present studies to determine the types of particle arrangements obtained in packings; this technique is apparently novel. Then, by using cylindrical containers of different diameters, packing densities were obtained which were compared with calculated expectations based on the packing arrangements discovered. The manner in which spheres arranged themselves in packings will be described first.

III. Packing Arrangements of One-Size Spheres

Containers with a cross-sectional shape that had corners, for example, square, when packed with one-size spheres caused a predominately square pattern in the surface spheres viewed through the container wall, although numerous colonies of rhombic pattern and random arrangement were observable. Cylindrical containers caused almost a complete rhombic pattern.* The important fact is that approximately the same packing density was attained for all these packings. The obvious explanation for this circumstance is that the three-dimensional packing arrangement was the same, which should be apparent from the following discussion.

From solid geometric considerations it can be shown that one-size spheres can be systematically arranged in space in four different schemes, i.e., simple cubic, orthorhombic, double-nested, and close-packed, † which ideally pack to 53.36, 60.46, 69.81, and 74.05% of theoretical density, respectively.

* In all cases discussed, the effective container cross section was large compared with the sphere size.

† This terminology has been chosen because it seemed to be the most descriptive for the present purpose. Graton and Fraser (footnote 6) in an excellent review of geometric principles defined the four arrangements in strict crystallographic terminology as follows: simple cube, orthorhombic (two orientations), tetragonal-sphenoidal (or wedge-shaped tetrahedron), and rhombohedral (the two rhombohedral cases are the face-centered cube and the close-packed hexagon). White and Walton (footnote 1) designated the four arrangements: cubic, single stagger, double stagger, and pyramidal (for face-centered cubic) and tetrahedral (for hexagonal close-packed).

Table II. Packing of One-Size Spheres

Mesh size	Density of material (gm./cc.)	% of theoretical density*
7	7.52	62.5
20	7.52	62.5
60	7.58	62.5
400 (tungsten)	19.30	59.5
400 (aluminum)	2.70	61.0

* The expression "% of theoretical density" has been used ambiguously by authors. In strict use, it designates the degree to which a solid body approaches its maximum or crystal density. The term finds wide application in describing sintered products which typically have residual porosity within particles as well as between particles. Unlike sintering phenomena, the end point of mechanical packing is elimination of the voids between particles; the porosity of particles is unaffected by packing. Thus, usage for mechanical packing is to consider the actual density of the particulate material as the theoretical density.

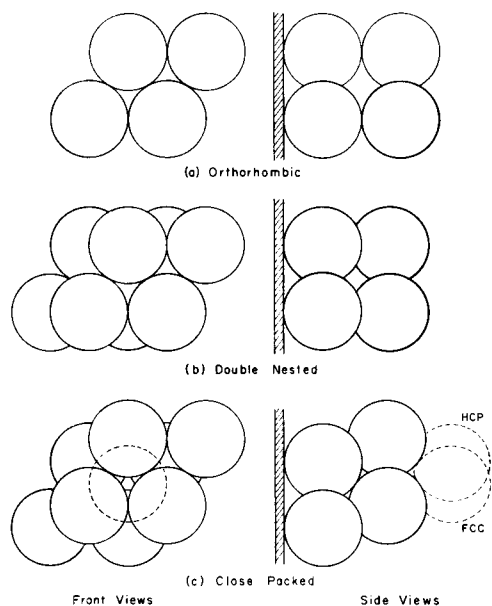


Fig. 1. Possible sphere arrangements in a packing tube.

Simple cubic is not pertinent to this discussion since such a packing is unstable and does not occur experimentally.

All single-sphere-size packings were found to consist almost entirely of the orthorhombic arrangement shown in Fig. 1 (a). Here the four spheres constituting the unit cell are depicted in the orientation they were observed to assume in the container. The container wall is shown in cross section in the side views of Fig. 1. The compression direction was vertical, causing an axial nesting of spheres and a tendency for transverse spreading of spheres. Thus, for a cylindrical container, layers of spheres consisting of concentric circles were observed. This experimentally observed arrangement will be described in detail shortly.

The double-nested arrangement, Fig. 1 (b), illustrates systematic nesting of spheres in both axial and transverse directions. As indicated earlier, an orthorhombic arrangement had a calculated packing density 60.46% of theoretical density and a double-nested arrangement a calculated packing density 69.81% of theoretical density. Because measured packing densities averaged about 62.5% of theoretical density, it is probable that only about a fifth of possible double-nesting actually occurred during packing.

Triple-nesting, which is illustrated in Fig. 1 (c), produces the closest possible packing, either face-centered cubic or hexagonal close-packed. Such close-packed arrangements cannot occur in mechanical packings because the vertical force would cause, for example, the two elevated spheres (side view of Fig. 1 (c)) to move downward to a layer position identical to the double-nested case. Thus, for the practical case of a vertical vibratory translation of a bed of spheres, packing tends to produce nesting of spheres in the vertical direction and lateral spreading of spheres, thereby stabilizing the orthorhombic single-nested arrangement.

IV. Effect of Container on Packing Efficiency of One-Size Spheres

The size of the container has a pronounced effect on packing density. Figure 2 shows experimental and calculated packing density data for arrangements in which the ratio of the container diameter, D , to the sphere diameter, d , varied

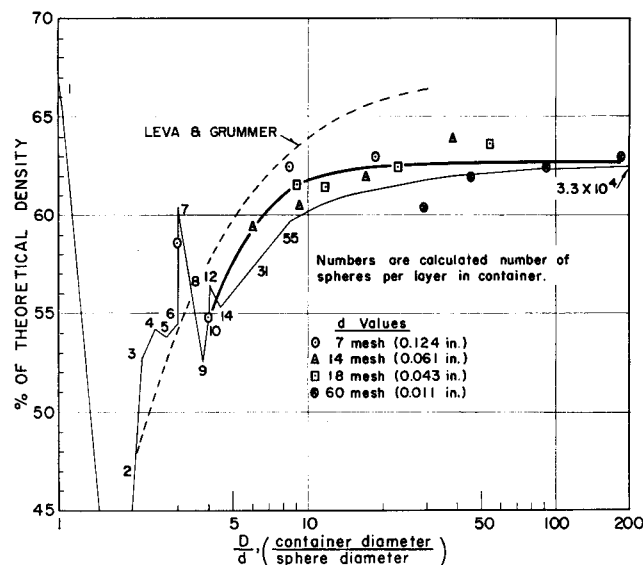


Fig. 2. Effect of container size on efficiency of packing of one-size spheres.

from 1:1 to about 200:1. The first condition was, of course, a column of spheres fitted in a tube of the same diameter. As the container increased in diameter, the density of the packing first decreased and then increased as 2, 3, or 4 spheres formed a layer. Only values for an integral number of spheres have been considered. All calculations were supplemented by geometrical construction and analysis to ascertain the optimum arrangement. Seven spheres produced the familiar hexagonal layer which is associated with efficient packing. Pertinent data for the calculated values are presented in Table III. For 9, 10, and 12 spheres in a layer, a ring of spheres was formed around a center of 3 spheres.

At 14 spheres per layer, the orthorhombic ring pattern was started where rings of spheres formed around a center of 4 spheres. Only the data for complete filling of rings of 14, 31, 55, etc., have been tabulated and plotted. These filled rings did not, however, cause a tight or locked arrangement. There was room for some of the spheres to move in a lateral direction, which evidently resulted in periodic entry of extra spheres to layers. This orthorhombic arrangement can be described by an empirical equation based on experimental data.

$$S = 4(r + 1) + \frac{13}{2} \sum_{i=0}^r r \quad (1)$$

S = total spheres per layer when $S \geq 14$.
 r = number of rings in layer.

Thus, if M = total spheres in the packing, h = total height of the packing, and $\sum_{i=0}^r r = \frac{r}{2}(r + 1)$, then

$$M = S \frac{2h}{\sqrt{3}d} = \left(\frac{13}{4} r^2 + \frac{29}{4} r + 4 \right) \frac{2h}{\sqrt{3}d} \quad (2)$$

At $D/d = 200$ there were about 33,000 spheres per layer and packing density reached a value of 62.5% of theoretical density, which agreed very closely with experimental maximum values.

For practical purposes, the curve indicates that maximum packing density can be obtained if the container diameter

Table III. Calculated Parameters of Mechanically Packed Sphere Arrangements

Spheres per layer	D/d	Layer height times d	% of theoretical density	Calculated number of spheres in rings of each layer				
Optimum arrangements of spheres								
1	1.00	1.000	66.7	1				
2	2.00	0.707	47.0	2				
3	2.15	0.817	52.7	3				
4	2.41	0.842	54.2	4				
5	2.68	0.849	53.8	5				
6	3.00	0.814	54.6	5	1			
7	3.00	0.855	60.5	6	1			
8	3.30	0.859	56.8	7	1			
9	3.75	0.860	52.5	6	3			
10	4.00	0.758	54.8	7	3			
12*	4.04	0.866	56.4	9	3			
Orthorhombic arrangement of spheres								
14	4.41	0.866	55.2	10	4			
31	6.41	0.866	57.9	17	10	4		
55	8.41	0.866	59.7	24	17	10	4	
$\frac{13r^2}{4} + \frac{29r}{4} + 4$ where r = number of rings in layer around 4 center spheres	∞	0.866	62.5	31	24	17	10	4

* 11 and 13 cannot occur since 12 and 14 occur at smaller D/d .

is at least an order of magnitude larger than the sphere diameter.

To obtain the experimental points, tubing $\frac{3}{8}$, $\frac{1}{2}$, 1, $1\frac{1}{2}$, and 2 in. in diameter was used for packing 7-, 14-, 18-, and 60-mesh steel shot. It is interesting to note that the 7-mesh shot could be packed to 58.6% of theoretical density in a $\frac{3}{8}$ -in. tube where a hexagonal layer arrangement occurred, but to only 54.9% of theoretical density in a $\frac{1}{2}$ -in. tube.

The data of Leva and Grummer³ are also presented in Fig. 2 for glass, steel, and porcelain spheres having discrete diameters between 0.172 and 0.730 in. and packed in metal tubes $\frac{3}{4}$ to 4 in. in diameter. The tubes were pounded with a hammer to produce packing. The data were best fitted by a straight line on linear rectangular coordinates so it is evident that the present data do not fall on a straight line on such coordinates. The packing densities they report are slightly higher than present values and those of other investigators.^{1,11} Their data give no indication of the inverse density relation at $D/d = 3$ and 4 demonstrated by the present investigation. Nevertheless, there is satisfactory general agreement among the data.

V. Binary Packings

A packing-density diagram is plotted in Fig. 3 for binary mixtures of coarse spheres (7-mesh) with six different sizes of fine spheres. Obviously, the smaller the diameter of the fine spheres, the greater was the density of the packing. Qualitatively, this behavior would be expected since the finest material was best able to fill the interstices of the coarse material. However, it is significant that these data could be obtained by only one experimental technique. The coarse component by itself was vibrated until it attained a minimum volume. Vibration was stopped, and the fine component was poured into the container and vibration continued until a minimum volume was again attained. When making ternary and quaternary packings, the same procedure was used for each successive component. Thus, the coarse component acted as a fixed* porous filter bed which was filled by the fine component.

* The phenomenon of dilatancy must also occur (see J. J. Hermans, Flow Properties of Disperse Systems. Interscience Publishers, Inc., New York, 1953. 450 pp.). A rigid bed of spheres subjected to vibration must separate slightly before they can slide one over another.

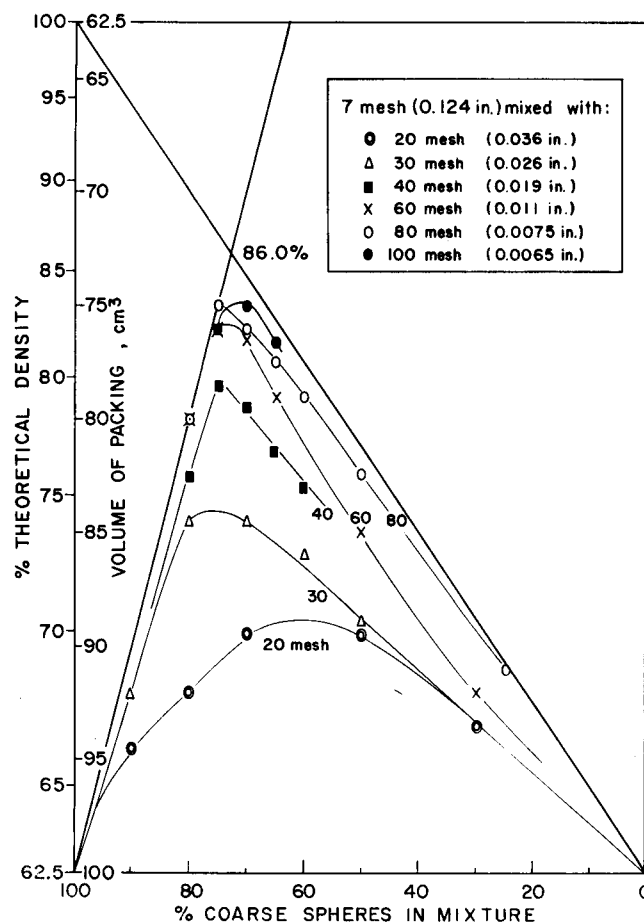


Fig. 3. Binary mechanical packing of coarse steel shot with some other sizes.

Inferior and lower final density packing resulted when the components were blended, poured into the container, and vibrated. In fact, effective blending could not be achieved when components varied considerably in size. For example,

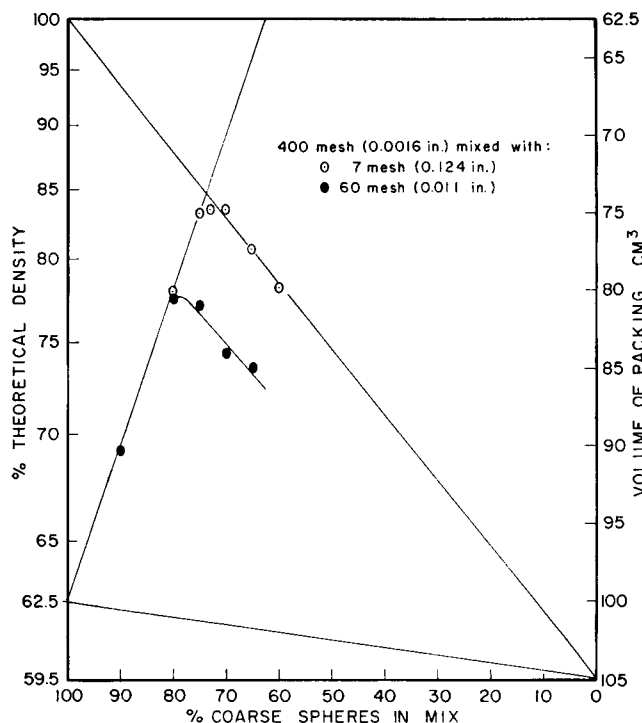


Fig. 4. Binary mechanical packing of coarse spheres with 400-mesh tungsten spheres.

7-mesh and 60-mesh shot showed gross segregation after attempted blending. It is likely that with sufficient time and intensity of vibration homogeneity could be achieved in a blended but segregated mixture.

It may be noted that the curve for 20-mesh shot has a bell shape, unlike the other curves. The 20-mesh shot was found to be too large in diameter to filter through the bed of 7-mesh shot. Ordinarily, a weighted plunger rests on the bed of material during a run. In the present case, the weight

on the material was reduced to the point where the 7-mesh bed expanded slightly during vibration, thus increasing the effective pore size of the bed. It was then possible for the finer material to filter through the bed of coarse material. However, the systematic packing of the coarse shot was lost, so that an intimate mixture would be a more apt description for this condition than would a packing.

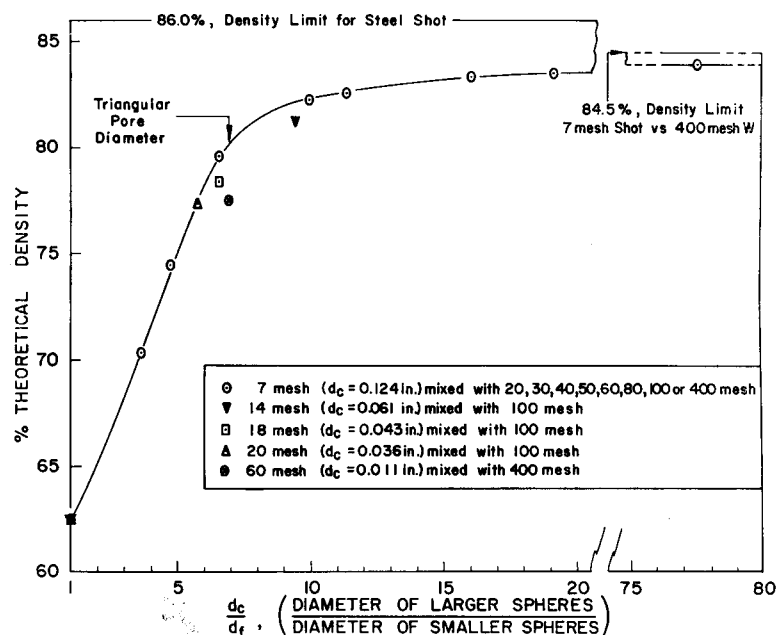
Figure 4 shows the results for 400-mesh tungsten which packed to only 59.5% of theoretical density mixed with either 7-mesh or 60-mesh steel shot which packed to 62.5% of theoretical density. These results could not be plotted on Fig. 3 because the limiting density was lower (84.7% vs. 86.0%) and the optimum composition was different, i.e., 74.0% vs. 72.3% coarse component. Figure 5 demonstrates the close correspondence of the data to the limiting case when there was a great difference (77.5 times) in the sizes of coarse and fine.

In Fig. 5 are plotted the maximum density values for each of the experimental curves of Figs. 3 and 4. Although the curve was drawn for the 7-mesh shot as the coarse component, it is apparent that other combinations of coarse and fine fell close to the curve, indicating that the size ratio of spheres governed the packing efficiency and that their absolute size was relatively unimportant in the size range studied.

The knee of the curve of Fig. 5 occurs at about $d_c/d_f = 7$. Below this value packing efficiency decreased rapidly. This value corresponds closely to the triangular pore path formed by any three close-packed spheres of equal size. The pore path effective diameter, p_t , was equal to the largest diameter sphere that could pass through this opening. If d_c is the diameter of the three spheres forming the triangular pore, then $p_t = d_c(2/\sqrt{3} - 1)$ or $0.154d_c$. The triangular pore was smaller than the square pore, $p_s = d_c(\sqrt{2} - 1)$ or $0.414d_c$, or than any intermediate-size pore; it therefore governed movement of the fine spheres through the stationary packing of coarse spheres.

It was shown in the previous section that an orthorhombic one-size packing arrangement prevailed experimentally. In this arrangement (refer to Fig. 1) there were square pore passageways running the length of the packing separated by concentric "tubes" of spheres with triangular pore holes in them. Full-length packing, although inefficient, could therefore occur even though $d_c/d_f < 7$.

Fig. 5. Maximum observed packing of some binary mechanical mixtures of spheres.



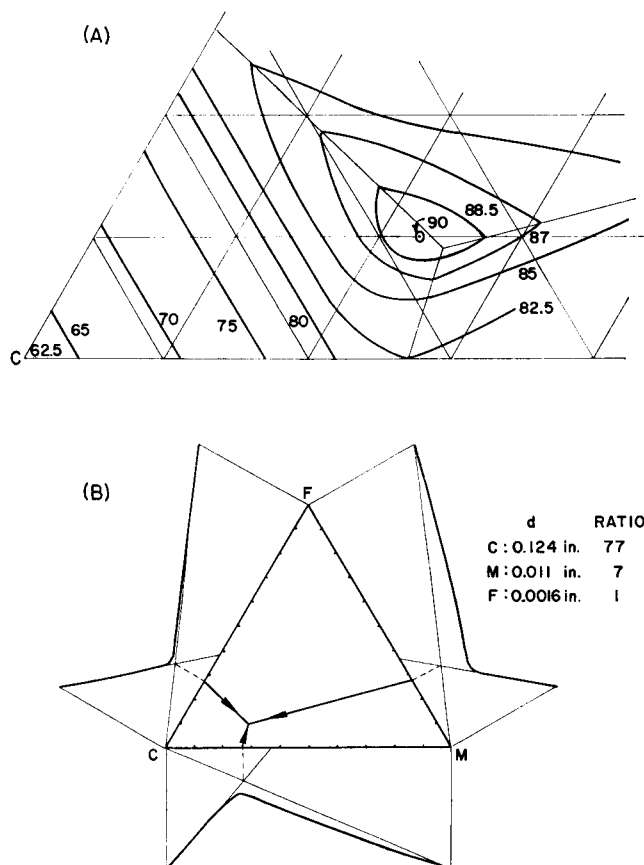


Fig. 6. (A) Experimental detail of coarse-sphere-size corner showing contours of per cent theoretical density and (B) concentration triangle for mechanical mixture of three sizes of spheres (coarse, medium, and fine).

VI. Ternary Packings

Ternary packings are analogous to equilibrium in a three-phase system and can be represented by a concentration triangle such as that in lower Fig. 6. The experimental binary equilibrium diagrams for coarse-medium, coarse-fine, and medium-fine are reproduced. Following standard geometric construction, the intersections of the three planes

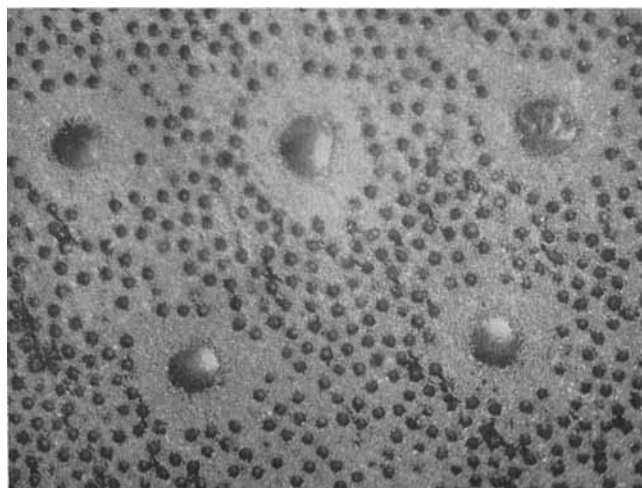


Fig. 7. Ternary packing of spheres to 90.0% of theoretical density with composition of 67% of 7 mesh (0.124 in. in diameter), 23% of 60 mesh (0.011 in. in diameter), 10% of 400 mesh (0.0016 in. in diameter). (×11.)

for the ideal limiting case are the three lines identified by arrows and joining at the point of maximum packing density. Maximum density occurred at volume percentages 66:25:9 for coarse, medium, and fine components in size ratio 77:7:1 (7, 60, 400 mesh, respectively). Experimental results for this packing, shown in upper Fig. 6, indicated a maximum density 90.0% of theoretical density at the 67:23:10 composition. The density relations can be computed as follows:

$$\% \text{ theoretical density} = 100 \left[1 - \frac{v_n X_n}{v_n X_n + V} \right] \quad (3)$$

n = number of components in packing.

v_n = volume fraction of voids in finest component.

X_n = volume fraction of finest component, or $v_{(n-1)} X_{(n-1)}$ for $n \geq 3$.

V = a constant; volume of solid material.

As shown in Table II, coarse and medium packed individually to 62.5% of theoretical density (37.5% voids) and the 400-mesh fine component packed to 59.5% of theoretical density (40.5% voids). From Fig. 3 the densest packing of coarse and medium occurred at a composition 72.7% coarse and 27.3% medium. Then, for the experimental conditions cited, the limiting, or the densest ternary packing that could be expected would be, using equation (3):

$$\% \text{ theoretical density} = 100 \times \left[1 - \frac{\frac{40.5 \times 62.5}{59.5} (37.5 \times 0.273)}{\frac{40.5 \times 62.5}{59.5} (37.5 \times 0.273) + 62.5} \right] = 100 - 6.5 = 93.5$$

Thus, for the test cited, a packing density 90.0/93.5 or 96.5% of this limiting density was attained experimentally.

Figure 7 shows a photograph of a typical region of a ternary packing of 7-, 60-, and 400-mesh spheres. The glass packing container was 1½ in. in diameter, and the exposure was perpendicular to its surface. The rhombic stacking arrangement of the layers of coarse spheres is apparent. The medium component gave indication of rhombic arrangement, but any regular arrangement was disrupted by the coarse spheres. Calculations compared with measurements taken from the photograph indicated that both the medium and the fine spheres went as far as possible into the wedge-shape void between the sphere surface and inside container wall and thus created the bull's-eye effect apparent in Fig. 7. This was a high-density packing (90% of theoretical density) and showed very few "holes" where fines were absent.

VII. Rate of Ternary Packing

Packing of medium spheres in a fixed bed of coarse spheres occurred rapidly and completely with relatively little dependence on the intensity or frequency of mechanical vibration or the weight bearing on the bed of material. Packing of the fine spheres in this binary packing occurred at a comparatively slower rate, and the rate of packing did depend on the intensity and frequency of the vibration. Evidently, the time required to produce a packing is proportional to the number of particles to be packed. In a 67% coarse:23% medium:10% fine volume concentration mixture, the ratio of diameters was 77:7:1, but the ratio of number of spheres was 1:490:69,000, respectively. Although only 10% of the solid material of the packing was in the form of fine spheres, there were about 142 times more fine spheres than medium spheres that had to migrate to proper locations.

The rate of entry of the fine component into a prepared binary packing was measured. Data for a constant 5-g (vector, or $\sqrt{2}$ r.m.s.) vibration acceleration and a variety of frequencies are plotted in Fig. 8. The ternary composition used was again the 67% coarse:23% medium:10% fine described above. For each run, a packing of 7-mesh shot was formed and then a binary packing 1 in. in diameter and 5 in. long was prepared after the addition of 60-mesh shot. The 400-mesh tungsten was poured onto this packing, the

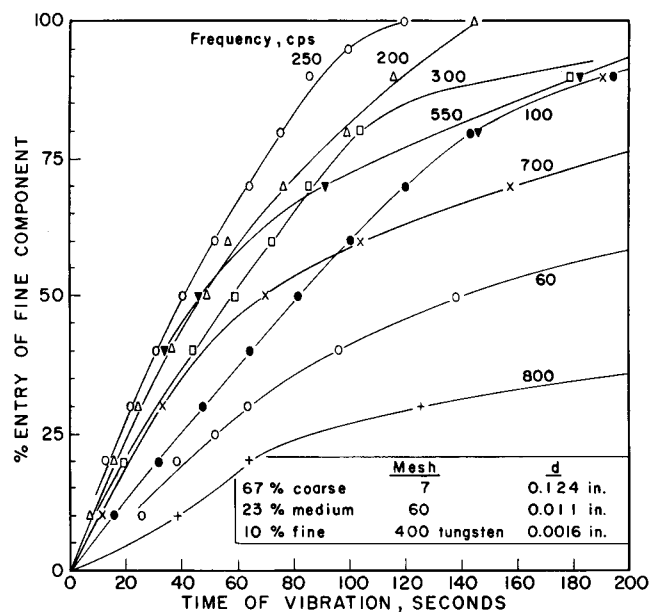


Fig. 8. Rate of entry of fine component into the prepared binary packing at acceleration of 5 g and variable frequency.

plunger replaced, and vibration continued while timing the volume decrease as the fines entered the packing.* Fastest packing occurred at about 250 c.p.s. vibration with a characteristic abrupt end point. At some higher and lower frequencies, the curves of Fig. 8 indicate that the rate of packing to about 50% completion was almost as rapid as for 250 c.p.s., but thereafter the rate of packing slowed continuously. Figure 9 (C) shows packing at 250 c.p.s. in which about 60%

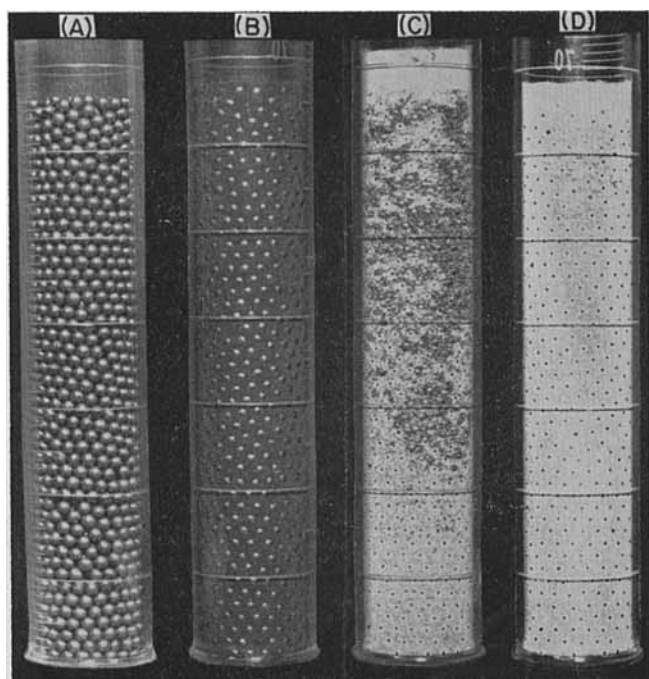


Fig. 9. Formation of a ternary packing in a 100-ml. graduate. Composition of the packing is 67% 7-mesh steel shot, 23% 60-mesh steel shot, and 10% 400-mesh tungsten shot. (A) Coarse component packed to 62.5% of theoretical density, (B) binary packing to 82.5% of theoretical density, (C) about 60% of fine component in binary packing, and (D) ternary packing of 90.0% of theoretical density.

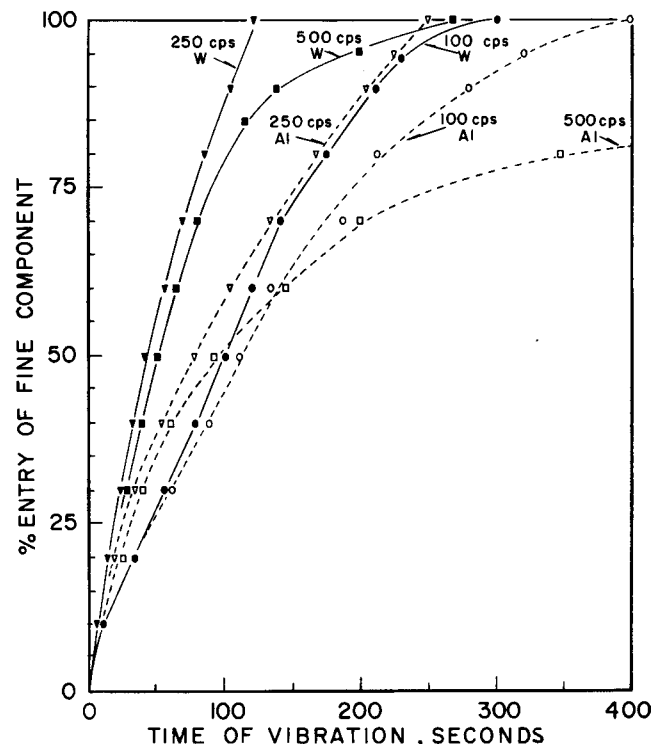


Fig. 10. Rate of entry of fine component of tungsten or aluminum into the prepared binary packing at acceleration of 10 g and variable frequency.

of the third component had entered the binary packing. Note the progress of filling of the interstices from the bottom upward without segregation of fine material. The patchwork appearance in the upper regions of the packing resulted from gradients in the tightness of packing for flow of the fine particles. Finally, Fig. 9 (D) shows completion of packing of the Fig. 9 (C) run. The density was 90.0% of theoretical density and could be reproduced repeatedly to $\pm 0.05\%$ with this material at different frequencies and accelerations. Note that the distribution of fines was uniform along the length of the packing except for a slight gradient near the top.

Figure 10 shows packing-rate curves at 10-g acceleration for tungsten and aluminum at three different frequencies of vibration. Again, 250 c.p.s. was best for both metals. Aluminum packed at a slower rate than tungsten at all frequencies and in addition showed a slower packing rate than did tungsten when near the finish of packing at high and low frequencies. The data of Figs. 8 and 10 are summarized in Fig. 11 where the percentage of fines that entered the binary packing after 2 minutes of vibration is plotted for various frequencies. Higher acceleration of the packings did speed up the packing of the fines at all frequencies for both metals, but it produced more of an increase for aluminum than for tungsten. The two powder materials appeared to be similar in shape and smoothness, but tungsten is more than seven times denser than aluminum. This density difference must, therefore, account for a slower rate of packing of the aluminum powder and the more pronounced effect

* Micrographic 16-mm. motion pictures taken through the glass-walled container during vibration show the details of packing phenomena. These may be borrowed by applying to the Nuclear Materials and Equipment Corporation, Apollo, Pennsylvania.

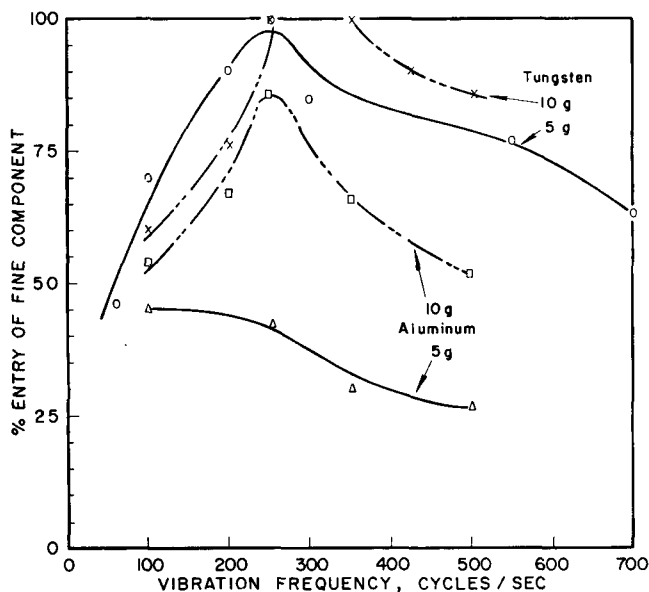


Fig. 11. Entry of fine component into the prepared binary packing after 2 minutes of vibration at acceleration of 5 or 10 g.

of acceleration. Again, it should be pointed out that the maximum packing density was not improved by changes in the rate at which the ternary packing was made.

VIII. Effect of Evacuation of Container During Packing

One report¹ cited previously states that "keeping the packing tube under vacuum of 70 to 70.5 cm. Hg materially assists packing." No additional information is offered, but it does seem logical that removal of the air from the

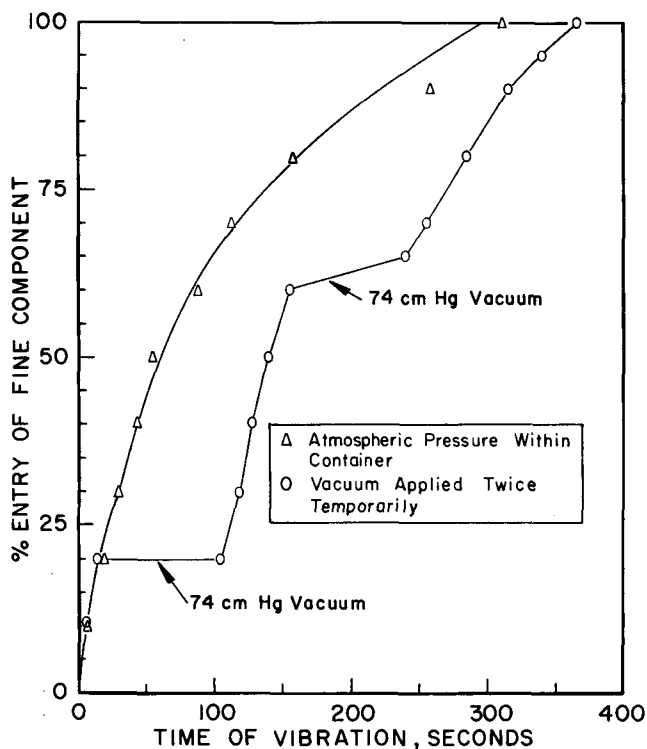


Fig. 12. Effect of vacuum on rate of entry of fine component into the prepared binary packing.

binary packing should speed the introduction of fines into it. Accordingly, experiments were designed to measure this effect quantitatively. A standard container was fitted with a vacuum nipple at its bottom and a close-fitting leather disk was placed on top of the packing under the plunger. A vacuum of 73 to 74 cm. Hg was quickly attained with a mechanical pump and was readily maintained during packing. The 67% coarse:23% medium:10% fine (400-mesh tungsten) composition was used.

The experimental results obtained were completely unexpected. No significant migration of fines into the binary packing was observed during a half-hour vibration at 250 c.p.s. and 5-g acceleration. A run under identical conditions, except without the vacuum pump running, was completely packed in about 5 minutes. Figure 12 shows a curve for this run and also a normal packing run which was interrupted twice; each time a vacuum was applied and the run continued for an arbitrary period of time during which no packing occurred.

Observation of the phenomena through the transparent wall of the container revealed that the fines were merely circulating in the upper regions of the container. Apparently, the air leakage around the leather gasket produced general turbulence and updraft currents. Streams of fines that had migrated some distance into the packing were observed moving upward until some clogging of the region occurred where-upon the upstreaming shifted to a more favorable locale. This phenomenon is ascribed to constant shifting of the least resistance path for air flow in the bed with infiltration of small amounts of fines.

An airtight plunger would be the next logical experimental approach. But having revealed the apparent aerodynamic complexity of the vacuum effect, additional experiments were deferred in favor of seemingly more practical research.

IX. Comparison of Electronic and Pneumatic Vibratory Equipment

For this study the only mechanical force used to effect packing was approximately sinusoidal axial vibration of the packed columns. Oscilloscopic examination of wave forms showed some distortion and superimposed harmonics, but it is unlikely that these conditions greatly influenced packing. Both electronic and pneumatic vibration equipment was used: a 10-lb. force vibration exciter; control console; VTVM calibrated for readings of displacement, velocity, acceleration; vibration pickups; and a Navco BH 1¹/₄ pneumatic piston bin rapper.

Pneumatic equipment is limited to low frequencies where it has been shown that packing rate is slower for a given acceleration level. This deficiency is mitigated, however, by characteristically high acceleration levels. Frequency and acceleration cannot be controlled independently since an increase in air pressure against the piston increases both. Electronic equipment operates over a wider frequency range so that an optimum or a resonant frequency can be selected for packing.

As shown in Fig. 13, the rate of producing a 67% coarse: 23% medium: 10% fine (400-mesh tungsten) ternary packing appeared to be comparable for both types of equipment operated at 38 c.p.s. and 2.5-g acceleration. At an air pressure of 40 lb. per sq. in. (56 c.p.s., 21 g) the fines packed in the prepared binary packing at almost a linear rate and a total time of 100 seconds. The pneumatic piston will operate at 70 lb. per sq. in. at 75 c.p.s. and 38 g, but the final packing will be no denser than for lower energy levels if no plastic deformation of material occurs. At high energy levels, whipping and nonaxial beats occur which cause drifting, i.e., a migration of fines across the container, thus producing a transverse concentration gradient of fines. Even a slight amount of drifting was discernible on close observation of a packing through its

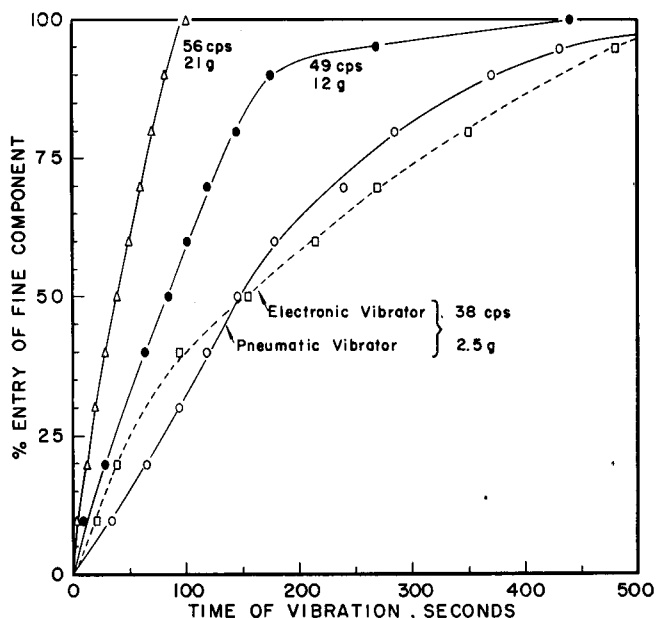


Fig. 13. Rate of entry of fine component into the prepared binary packing using a pneumatic piston vibrator.

transparent container. It was found that drifting was overcome by rotation of the container during the vibration.

Pneumatic vibrators are insignificant in cost compared with electronic equipment. It is true that they do not operate in the optimum frequency range for packing and are extremely noisy, but for repetitive industrial packing processes, they will undoubtedly find wide use. For research work, the greater versatility of electronic equipment is needed.

X. Quaternary Packings

Packings with four or five or more components followed the same relations just demonstrated for ternary packings. But, from the practical standpoint, it became more difficult to obtain materials with an adequate difference between particle sizes of the several components. Furthermore, the benefit of each additional component in increasing packing density was less than half that of the preceding component. Multicomponent packings could, however, be useful in producing homogeneous packed beds or columns with a dilute distribution of a specific fine material. A number of interesting possibilities present themselves: The composition of the fines, continuously added, could be altered so that similar gradations would be produced and maintained along the length of the packing; the fines of a fourth or even fifth component could be a fluxing material intended for bonding of the packing with a subsequent treatment, etc. Thus, although components beyond three did not significantly contribute to high packed density, they were useful for uniform distribution of a small amount of another very fine material throughout the packing.

Glass spheres were used as the coarsest component for quaternary packing experiments. A second component was then added periodically and vibration was continued until no further volume shrinkage resulted and the height level of both components was the same. This procedure was repeated for the last two components. The volume of each component added was then measured after pouring out and sieving the material.

Reference to Fig. 2 indicates for $D/d = 3$ that seven spheres forming a hexagonal layer arrangement produced a packed density 60.5% of theoretical density. The glass spheres $1/2$ in. in diameter were therefore packed in a tube $1 1/2$ in. in inside diameter where a packing 58.0% of theoretical density was obtained. Results of packing the glass spheres, 14- and 60-mesh shot, and 400-mesh tungsten are presented in Table IV. The experiment was repeated twice and results for the quaternary packing were reproducible to $95.1 \pm 0.3\%$. The limiting density values were determined using equation (3).

Note that the d_c/d_f ratio for the binary packing was 8.3, which is just above the knee of the packing efficiency curve in Fig. 5, whereas the ratio between components 2 and 3 was 5.4 and between components 3 and 4 was 7.0, both of which were at or below the "triangular pore diameter" at the knee of the curve. Surprisingly, the largest discrepancy between limiting and experimental values (5.9 percentage points) occurred for the binary packing.

The experimental packing density of 95.1% of theoretical density for a quaternary packing must be considered as remarkable since it was equivalent to a pressed and high-fired ceramic. Furthermore, the products were entirely different; a ceramic is, of course, a solid hard body, whereas the packing had no green strength and could actually be poured from its container.

Use of spherical particles as large as $1/2$ in. for packing is pertinent for some applications but is not for the fabrication of most objects. It is interesting to note size ranges required for a quaternary packing if the components shown in Table IV are simply scaled down. Using 10μ powder as the fine, the following combination would be required: 10 μ , 200 mesh, 45 mesh, 7 mesh. However, the micronized powder would probably pack poorly by itself and would be so agglomerated that its effective particle size would be somewhat larger.

It might be assumed that using glass or plastic containers would impose a serious limitation on the experimental range of force or pack loading required for packing. This did not appear to be the case. In the present work, it was found that only sufficient dead load on the bed of material was required to prevent its expansion during vibration. For acceleration of materials up to 10 g as little as 1 lb. per sq. in. was adequate, although 8 lb. per sq. in. was generally used. Without this light load, the top surface of the bed of material "boiled" and a dilated arrangement of the spheres prevailed downward into the packing a distance depending on the acceleration. Loading of the bed of material over the range 1 to 16 lb. per sq. in. had no detectable influence on the rate of entry of the finest component into a prepared packing, or on the ultimate density of such packings. This was a logical expectation (at least for hard spherical material) because, to reiterate, no plastic deformation or interlocking of particles occurred, which was

Table IV. Density of a Quaternary Packing

Component, n	Mesh	d	d ratio	v_n	Composition				% of theoretical density	
					X_1	X_2	X_3	X_4	Calculated limit	Experimental
1		0.505	316	45.3	1.00				60.5	58.0
2	14	0.061	38	37.5	0.726	0.274			85.9	80.0
3	60	0.011	7	37.5	0.647	0.244	0.109		94.2	89.8
4	400	0.0016	1	42.5	0.607	0.230	0.102	0.061	97.5	95.1

proved by the fact that the packed material could be poured from its container.

XI. Summary

(1) Spherical particles of one size when mechanically packed in large containers were observed to be arranged in an orthorhombic pattern. For square or cylindrical containers the packed density was normally about 62.5% of theoretical density, although the very fine components packed less efficiently. When the container diameter was less than about ten times the diameter of the one-size spheres being packed, it severely decreased the efficiency of packing and pack density; calculated and experimental quantitative relations have been presented.

(2) Only one specific packing procedure was found to be successful in producing high-density, homogeneous, multi-component packings; the preceding component or components acted as a stationary filter bed for the next finer component which was placed on top of the bed and the whole assembly was mechanically vibrated to effect entry of the fines into the bed. Thus, packing was accomplished in a number of steps equal to the number of components to be packed.

(3) To produce efficient packing, there should be at least a sevenfold difference between sphere diameters of the various individual components. This size difference was shown to be associated with the "triangular pore size" in the established packing through which the next component had to migrate to reach a permanent site.

(4) A four-component packing with 95.1% of theoretical density was prepared from spheres with diameter ratios 1:7:38:316 and volume compositions 6.1%:10.2%:23.0%:60.7%, respectively. The calculated limiting density for this packing based on infinite difference in size between components was 97.5% of theoretical density.

(5) The ultimate density of the packing was essentially independent of the rate at which it was formed. Affecting the rate of packing were frequency and acceleration of mechanical vibration, density of materials, fineness of components (number of particles per unit volume), and evacuation of container during packing. Not affecting the rate of packing were dead load on the column of material during packing (within the load range studied) and type of vibratory equipment used, i.e., pneumatic or electronic, provided the energy inputs, as determined by frequency and acceleration, were similar.

(6) No plastic deformation was involved in packing. All packing, regardless of the number of components used or the intensity or duration of vibration, could be poured from the glass graduates used as containers for packing. Hence, packing is simply an efficient arrangement of spheres of prescribed sizes and proportions and arranged by a stepwise process.

Acknowledgment

Thomas Connor has been responsible for performing most of the experiments and for collection of the data.

VI International Congress on Glass

Sheraton-Park Hotel, Washington, D. C., U.S.A.

July 8-14, 1962

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